

A Conditional Numerical Acceptance Certificate for the MQM Thermal Boundary Latch

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Abstract—We evaluate the complete finite one-photon response of a 192-mode flat-band Gaussian reservoir using short-step Taylor propagation. Directed quadrature-parameter intervals and explicit binary64 operation-count bounds give a conditional numerical acceptance certificate under declared arithmetic assumptions. Combining this finite response with the inherited vacuum, covariance and Gaussian remainder allowances yields a fixed-launch retained thermal error below $773/2^{26} + 2^{-128} < 2^{-16}$ at unchanged $\beta = 5/16$ and $T = 16$. The result does not establish permanent retention, arbitrary-input diamond error, native admission, a physical reservoir implementation, or quantum advantage. A separate exact spectral-carrier construction narrows one capacity-selection question without deriving $7/768$ as a physical coefficient.

Index Terms—Quantum control, thermal reservoir, numerical error bounds, finite excitation sectors.

I. FROZEN CONTRACT AND EVIDENCE LEVEL

Keep $\Omega = 64$, $B = 16$, $\kappa = 8$, $\beta = 5/16$, $T = 16$, $h = g = 1$, and $\lambda^2 = 4\sqrt{2}$. The source retained system has charge multiplicities $(1, 4, 3)$. The launch is $p0$ with arbitrary identity spectators. The finite stationary Gauss–Legendre bath has $M = 192 = 2^6 \cdot 3$ mathematical modes. It requires $D_2 = 3 + 4M + M(M + 1)/2 = 19299$ two-excitation coordinates, not physical qubits.

The inherited exact finite response identity is [1]

$$\Delta_2^M = \mathcal{R}_{T,2}(|p0\rangle\langle p0| \otimes \Gamma_1(N)) - (\text{tr } N)\rho_0^M(T). \quad (1)$$

All 192 covariance-weighted columns are propagated. This response derivative is not a photon-cutoff approximation. The prior finite response was exposed before this engineering certificate; neither this result nor the adjacent spectral proposal is an independent prediction.

The continuum bridge retains the conditional centered vacuum-response allowance 2^{-30} [2] and separate covariance allowance 2^{-57} [3]. The inherited cold ceiling is $3/2^{18} + 2^{-128}$ and fourth-and-higher Gaussian corrections are below 2^{-25} [1]. These dependencies are assumed; this release does not replace their external review.

II. DIRECTED PARAMETER ENCLOSURES

For normalized bath measure $d\delta/(2B)$, take exact Legendre roots x_j , standard weights w_j , frequencies $\delta_j = 16x_j$, couplings $c_j = \sqrt{64w_j/\pi}$, and occupations

$$n_j = (e^{(5/16)(64+\delta_j)} - 1)^{-1}. \quad (2)$$

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Each exact root is enclosed by a disjoint sign-changing interval of radius 2^{-44} around its stored binary64 node. There are 192 intervals and exactly 192 simple roots. Directed 200-digit Decimal recurrence verifies the signs. The global bound

$$\sup_{[-1,1]} |P''_{192}| \leq \frac{191 \cdot 192 \cdot 193 \cdot 194}{8} \quad (3)$$

encloses derivative variation from the bracket center. The exact weight formula $w_j = 2/[(1 - x_j^2)P'_{192}(x_j)^2]$ is then evaluated in intervals. Machin alternating rational sums enclose π ; square roots and exponentials use correctly rounded Decimal evaluations expanded to adjacent representable endpoints.

The resulting parameter receipts give

$$\|c - \hat{c}\|_2 < 4.318 \times 10^{-6}, \quad (4)$$

$$d_H < 1.296 \times 10^{-5}, \quad (5)$$

$$\sum_j |n_j - \hat{n}_j| < 3.059 \times 10^{-18}, \quad (6)$$

$$\text{tr } N < 2^{-16}. \quad (7)$$

Decimal displays are diagnostic. Exact rational endpoints are retained in the replay. The Hamiltonian allowance uses $2 \max |\delta_j - \hat{\delta}_j| + 3\|c - \hat{c}\|_2 + |\lambda - \hat{\lambda}| + 2^{-40}$; the final term covers bosonic-factor and sparse-assembly rounding. The coefficient rounding bound is conservative, but adequate.

An initial helper allowed ambient 28-digit precision in negations. Its output is preserved and excluded from promotion. The repaired helper uses 200-digit context and exact sign reversal. The amendment changes no physical parameter or acceptance target.

III. PROPAGATION AND READOUT ARITHMETIC

Each stored real sparse row has at most 194 nonzeros, below the declared ceiling196. The absolute-value generator has norm below64 by the sector bath and collective-exchange bound [1]. Real and imaginary vectors are propagated separately. Each of128 steps has duration 1/8 and degree48:

$$v_{k+1} = v_k + \sum_{j=1}^{48} \frac{(-iH/8)^j}{j!} v_k. \quad (8)$$

The implementation computes terms recurrently and adds them successively. Binary64 round-to-nearest arithmetic, correctly rounded division and square root, and ordinary sequential or pairwise sparse accumulation are explicit assumptions. No accuracy of a floating eigensolver is required.

With $\gamma = 2^{-43}$, the real sparse product/division error is bounded normwise by $\gamma|H|$. Term recurrence errors sum to

at most $16\gamma e^8$; partial-sum errors fit an additional $16\gamma e^8$ allowance. Since $e^8 < 3^8$, a step allowance is

$$s = 32\gamma 3^8 + \frac{8^{49}}{49!(1-8/50)} + 2^{-800}. \quad (9)$$

The last term covers tiny/subnormal intermediate operations. Exact step propagation is unitary. With $128s < 10^{-5}$, telescoping and $(1+s)^{128}$ give total vector error at most $2 \cdot 128s < 2^{-17}$. A Duhamel charge $16d_H$ separately covers ideal-to-stored generator mismatch. Thus put $e = 16d_H + 2 \cdot 128s$.

For a normalized launch, vector error e changes its dyad by at most $e + e^2/2$ in half trace norm. The hot weighted mixture and cold subtraction together cost at most $(\text{tr } N)(2e + e^2)$. Occupation replacement costs $\sum |\delta n_j|$. Readout uses at most 192^2 products per retained entry; the conservative reduction allowance 2^{-31} , charge factors and $\text{tr } N < 2^{-16}$ fit a total debit 2^{-36} . The replay includes the detailed operation-count argument.

For the final stored matrix, define the exact binary-rational bound

$$b_{\text{mat}} = \frac{1}{2} \sum_{a,b} (|\Re \hat{\Delta}_{ab}| + |\Im \hat{\Delta}_{ab}|). \quad (10)$$

It bounds half trace norm by the triangle inequality for matrix units, without assuming eigenvalue accuracy.

IV. CONDITIONAL ACCEPTANCE RESULT

Theorem 1. *Under the declared arithmetic assumptions and inherited continuum/Gaussian bounds, the fixed-launch second-order response has an upper allowance below $3/2^{26}$, and the retained thermal error is below $773/2^{26} + 2^{-128} < 2^{-16}$.*

Proof. Exact fraction arithmetic sums b_{mat} , weighted propagation error, occupation error, 2^{-36} readout error, 2^{-30} vacuum comparison and 2^{-57} covariance comparison. The result is approximately 3.37334×10^{-8} and is strictly below $3/2^{26}$. Adding the source cold ceiling and higher-order remainder gives

$$E_\beta < \frac{3}{2^{18}} + 2^{-128} + \frac{1}{2^{25}} + \frac{3}{2^{26}} = \frac{773}{2^{26}} + 2^{-128} < 2^{-16}. \quad (11)$$

The accepted threshold is unchanged. Numerical enclosure assumptions and source theorem dependencies remain explicit. $\square$

The propagated eigenvalue diagnostic is $2.2505874657 \times 10^{-8}$ and agrees with the prior exponential-action estimate. This agreement is a simulation control; the acceptance argument uses the conservative exact-entry and error ledger.

V. SEPARATE CAPACITY-SELECTION AUDIT

On the original tetrahedral 15-coordinate chain register with 28 incidences, let D be its oriented boundary and $A = D - D^\top$. Exact arithmetic verifies $D^2 = 0$ and $A^3 = -4A$. For $H = -iA$, the projectors

$$P_0 = I + A^2/4, \quad P_+ = (-A^2 - 2iA)/8 \quad (12)$$

have ranks 1 and 7, are orthogonal, and commute with all 24 oriented S_4 actions. The proposed oriented spectral carrier

$P_0 + P_+$ has dimension 8 and a unique zero-mode reference. A uniformly mixed state on this carrier assigns active weight $7/8$.

This narrows the reference-selection question in a newly specified spectral construction. It does not identify that carrier with the previously declared three-bit Θ , derive uniform allocation, introduce 192 physical carrier coordinates, or justify division by 24 and 4. Those identifications, source dynamics, admissibility law and latch map remain OPEN. The exact identity $7/768 = (2^3 - 1)/(4 \cdot 4! \cdot 2^3)$ remains arithmetic provenance rather than a material constant.

VI. SCOPE AND NEXT GATE

The acceptance pass is conditional on software arithmetic and source reservoir assumptions. Independent external numerical/theorem audit remains OPEN. This fixed-launch point-time result does not establish permanent dark retention, finite-temperature hardware performance, arbitrary-input diamond error, full-word compilation, two-local native admission, or quantum advantage. Physical promotion remains zero. The next useful gate is an independently reviewed implementation certificate, followed by a bounded retention-window analysis.

REFERENCES

- [1] B. Mayes, *MQM Second-Order Specification and Frequency-Resolved Thermal Bounds*, development releases v0.1, 2026.
- [2] B. Mayes, *Time-Centered Gaussian Quadrature Embedding for the MQM Vacuum-Transfer Certificate*, v0.1, 2026.
- [3] B. Mayes, *MQM Chain Response*, v0.1, 2026.